

Master Implementation Plan & System Matrix: Pace Messenger Platform

Executive Overview & System Architecture

Pace Messenger is an enterprise-grade, multi-channel AI-enabled Customer Relationship Management (CRM) and Messaging Platform. The platform bridges real-time messaging (WhatsApp Cloud API, Live Chat, Web Push, Email, SMS) with advanced workflow automation, AI-driven customer intelligence, interactive event radar mapping, and automated campaign generation.

System Architecture Blueprint

```
[ Frontend: Angular 21 Single Page App ]
  |
  |— index.html (Leaflet Maps, FB SDK v18.0, App Host)
  |— App Routing & Standalone Components (24 Feature Modules)
  |— @foblex/flow (Visual Flow Builder & Canvas Engine)
  |— PrimeNG 21 + GSAP + Socket.io-client
  |
  ▼ (HTTP REST APIs / WebSocket Real-time Bi-directional Stream)
  |
[ Backend: Express 5 Node.js Server ]
  |
  |— Middleware (CORS, Helmet, Body Parsers, X-Project-Id Resolver)
  |— v1 Route Registry (73 Sub-route Modules)
  |— Services Layer (22 Services: AI Autopilot, Marketing Intelligence, Qdrant RAG)
  |— Real-time Engine (Socket.IO + BullMQ Redis Queues)
  |— Database Layer (Sequelize ORM - 84 Models - MySQL / SQLite)
  |
  ▼ (External Gateways & Integrations)
[ WhatsApp Cloud API ] [ Qdrant Vector DB ] [ AWS S3 / Firebase ] [ Meta Ads API ]
```

User Review Required

IMPORTANT

Key Architectural Decisions & Scope Highlights

1. **Tabular Matrix Format:** Every module from `index.html` down to internal backend services is detailed in the 6-column tabular overview below for direct side-by-side comparison.

2. **Visual Flow Execution Engine:** The UI utilizes `@foblex/flow` v19 for node-based visual drag-and-drop workflow editing. The backend execution engine for custom flow nodes provides runtime parity for decision branches, AI nodes, and automated webhook triggers.

3. **Multi-tenant Project Context:** Every incoming REST API request inspects `X-Project-Id` and `X-Environment-Mode` (Live vs Sandbox) headers to ensure tenant isolation across all 84 database models.

Master 6-Column Feature & Module Implementation Matrix

Phase #	Phase / Module Name & Page	UI File / API Controller / Model	Current Implemented Status	Pending / Incomplete Requirements	Advanced Possibilities (AI, Data & Process Innovation)
1	Public SaaS Landing Page & Marketing Portal	src/index.html features/landing/landing-view/landing-view.ts public/	Working: Responsive landing page with Hero sections, platform features, interactive multi-channel demo snippets, pricing tier breakdown, and contact forms. Loads Leaflet CSS/JS & Facebook SDK v18.0.	Dynamic customer testimonial carousel feed sourced from live customer reviews; multi-currency pricing switcher based on visitor IP geolocation.	AI Conversational Sales Bot: Interactive AI chatbot on <code>index.html</code> that answers SaaS capability questions, calculates ROI savings, and books live product demos automatically.
2	User Authentication & Session Security	features/auth/login-view signup-view forgot-password-view routes/user.route.js refreshtoken.route.js user.model.js	Working: Secure email/password login, JWT Bearer Token issuing, refresh token rotators, password reset request handlers, and login history logging (<code>loginhistory.model.js</code>).	Social Single Sign-On (SSO) via Google OAuth 2.0 and Microsoft Entra ID buttons; automated email verification token expiration workflow.	Zero-Trust Behavioral AI Security: Detects anomalous login locations/devices and dynamically prompts step-up MFA or throttles suspicious IP ranges automatically.
3	Multi-Tenant Organization & Workspace Selector	features/projects/projects-view routes/projects.route.js projects.model.js middleware/ (<code>X-Project-Id</code>)	Working: Project creation, organization workspace switcher UI, tenant header resolution via <code>X-Project-Id</code> and <code>X-Environment-Mode</code> (Live vs Sandbox).	Automated quota limit warnings when a workspace approaches allocated subscriber or message dispatches.	AI Tenant Health Monitor: Analyzes tenant activity patterns to predict workspace churn risk and trigger proactive CSM engagement alerts.
4	Executive Operational Dashboard	features/dashboard/dashboard.ts services/dashboard.service.js routes/customreports.route.js campaignmetrics.model.js	Working: KPI summary cards for total incoming messages, active chat conversations, campaign delivery rates, active agent count, and recent activity timelines.	Real-time auto-refresh on KPI widgets via WebSocket events without requiring manual page reload.	AI Daily Executive Briefing: Daily text/audio executive summary highlighting key channel spikes, lead conversion drop-offs, and high-priority escalation alerts.
5	Unified Multi-Channel Live Chat Inbox	features/live-chat/live-chat.ts routes/conversations.route.js routes/messages.route.js sockets/socket.service.js conversations.model.js	Working: Real-time bi-directional messaging UI (WhatsApp, Web Chat, Email, SMS), Socket.IO live stream, agent assignment dropdowns, canned response picker, and file attachment uploads.	Granular message search within specific conversation threads; typing indicator sync across multiple active agent views.	AI Copilot Live Assistant: Generates instant, context-aware reply drafts in the agent composer using RAG knowledge base snippets and sentiment analysis.

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6	Communication History & Audit Trail	features/history/history.ts routes/auditlogs.route.js activitylogs.route.js systemlogs.route.js auditlogs.model.js	Working: Complete historical log viewer for all incoming/outgoing message dispatches, system events, user activity trails, and audit logs with filterable table views.	One-click export of filtered communication history logs to encrypted CSV/Excel files for compliance auditing.	AI Compliance & Anomaly Detector: Scans historical audit logs to detect sensitive PII leaks, compliance policy breaches, or anomalous bulk data downloads.
7	Broadcast Campaign Manager & RFM Segmentation	features/campaigns/campaigns.ts create-campaign-view services/campaign.service.js rfm-segmentation.service.js campaigns.model.js	Working: Multi-channel broadcast creator (WhatsApp, Web Push, Email), target group selection, RFM (Recency, Frequency, Monetary) segmentation model, and scheduled dispatches.	A/B test variant message creator with automated winner auto-selection based on initial 10% sample open/read rates.	Predictive Campaign Optimizer: AI engine that forecasts campaign open/click rates before dispatch and recommends optimal sending times per audience segment.
8	Meta WhatsApp & Channel Template Manager	features/templates/templates.ts routes/messagetemplates.route.js messagetemplates.model.js integrations/whatsapp.api.js	Working: Meta-approved WhatsApp HSM template syncer, status checker (APPROVED, REJECTED, PENDING), dynamic header/body variable placeholders, and Email HTML template library.	Visual drag-and-drop HTML email template editor within the UI; real-time Meta template rejection reason parser.	AI Template Generator & Approver: Generates WhatsApp message copy pre-optimized to comply with Meta quality policies and avoid rejection algorithms.
9	CRM Contact Hub & Group Management	features/contacts/contacts.ts routes/whatsappcontacts.route.js contacttags.route.js optoutcontacts.route.js whatsappcontacts.model.js	Working: Contact profile directory, tag management, custom attribute fields, CSV/XLSX bulk importer (<code>xlsx</code> / <code>csv-parser</code>), contact group assignment, and opt-out/block list filters.	Automated contact deduplication engine based on fuzzy name and phone number matching.	AI Lead Intent & Profile Enrichment: Enriches contact profiles with public business data, scoring purchasing intent and predicting lifetime value (LTV).
10	Visual Workflow Engine & Drag-and-Drop Builder	features/flows/flows.ts flow-canvas-view/flow-canvas-view.ts @foblex/flow routes/flows.route.js flows.model.js	Working: Node-based visual canvas powered by <code>@foblex/flow</code> . Configures triggers, decision branches, delay timers, message dispatches, and webhook actions.	Complete backend execution interpreter for nested conditional logic and asynchronous delay steps (<code>scheduledtasks</code>).	Generative AI Flow Builder: Text-to-Flow generator that constructs complete visual automation workflows from natural language prompts (e.g., "Build a 3-step lead nurture flow").
11	Event Intelligence Radar (Geospatial Map)	features/events-radar/events-radar.ts services/events-aggregator.service.js events.model.js routes/events.route.js	Working: Interactive Leaflet geospatial map visualizing industry conferences, trade shows, and regional events. Category filters, event detail sidebars, and geo-pin clusters.	Radius-based spatial search to auto-select all CRM contacts located within X kilometers of an event pin.	Geo-Spatial AI Event Matcher: Automatically matches upcoming regional trade shows with nearby CRM leads and drafts localized WhatsApp meeting outreach campaigns.

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12	Market Intelligence Cloud	features/market-intelligence/market-intelligence.ts services/market-intelligence.service.js competitors.model.js marketdemandsignals.model.js	Working: Market demand signal tracking, city economic benchmarks (<code>cityeconomicbenchmarks.model.js</code>), competitor price tracking, and industry trend analytics.	Automated daily background web scrapers for competitor pricing updates and alert trigger thresholds.	AI Market Trend Forecaster: Machine learning model that predicts upcoming regional demand surges and alerts marketing teams to launch targeted campaigns early.
13	AI Marketing Intelligence Platform	features/ai-marketing-intelligence/ai-marketing-intelligence.ts services/ai-marketing-intelligence.service.js metaadintelligence.model.js aimarketingrecommendations.model.js	Working: Meta Ad creative intelligence tracking, ad fatigue scoring algorithms, competitor ad performance benchmarks, and automated AI marketing recommendation generation.	Direct 1-click apply action to implement AI recommendations directly into live Meta Ad campaigns.	Autonomous Creative Refresher: Detects creative ad fatigue in real time and automatically generates updated ad copy variations and image concepts.
14	Meta Ads Manager & Click-to-WhatsApp Lead Sync	features/ads-manager/ads-manager.ts routes/ads.route.js ads.model.js competitorads.model.js	Working: Meta Ad campaign performance dashboard, Click-to-WhatsApp ad tracking, ad spend monitoring, and lead conversion reporting.	Automated Webhook lead instant ingestion for Meta Lead Form ads directly into CRM contact groups.	AI Click-to-WhatsApp Ad Router: Predicts incoming ad lead intent and immediately routes high-value leads to top-performing live agents.
15	Payments Gateway & Collection Link Manager	features/payments/payments.ts routes/payments.route.js payments.model.js integrations/payments/	Working: Payment link generator, invoice tracking, payment status webhooks (paid, pending, failed), and customer transaction log history.	Multi-currency automated conversion rates and international payment gateway support (Razorpay, Stripe).	AI Smart Payment Recovery: Sends personalized WhatsApp payment reminders with dynamic discount incentives for early invoice settlements.
16	Integrations Gateway & Webhook Router	features/integrations/integrations.ts routes/integrations.route.js webhookendpoints.route.js webhooks/whatsapp.webhook.js	Working: Connector hub for WhatsApp Cloud API, Meta Business Manager, SMTP Credentials, Twilio SMS, Qdrant Vector DB, and custom webhook endpoint listeners.	Webhook payload replay and dead-letter queue (DLQ) viewer for debugging failed external webhook dispatches.	Self-Healing Webhook Monitor: AI monitor that detects third-party API rate limits or failures and automatically retries dispatches with exponential backoff.
17	Team Governance, RBAC & Workload Management	features/manage/manage.ts routes/role.route.js permission.route.js agentstatus.route.js agentworkloads.route.js	Working: Admin user management, role-based access control (RBAC), permission matrix configuration, agent online/offline status monitors, and workload balancing rules.	Granular field-level visibility controls (e.g., masking customer contact details for lower-tier agent roles).	AI Workload Equalizer: Dynamically re-assigns active conversations based on real-time agent response speeds, chat complexity, and queue depth.

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18	AI Autopilot Agents & Persona Configurator	features/agents/agents-view.ts services/agentic-autopilot.service.js aichatbot.service.js aipersonas.model.js aiinteractions.model.js	Working: AI persona manager (system prompts, tone of voice, knowledge scope), AI autopilot toggle per channel, and AI interaction logging (<code>aiinteractions</code>).	Automated sentiment threshold trigger to automatically escalate complex/frustrated customer chats from AI to Human Agent.	Autonomous Sentiment Handoff Engine: Real-time emotion analyzer that detects negative customer sentiment and seamlessly hands off conversations to live human agents.
19	Developer Portal & Embeddable SDK Generator	features/developer/developer-view.ts routes/apikeys.route.js widget.route.js public/sdk/ widgetconfigs.model.js	Working: API Key generation, secret key rotation, embeddable live-chat HTML/JS widget configurator (<code>public/sdk/</code>), and system diagnostic log view.	Interactive API sandbox testing console directly within the developer portal.	AI SDK Code Generator: Generates customized API integration code snippets in 6 programming languages (Node.js, Python, cURL, PHP, Java, Go) based on tenant settings.
20	User Account Profile & Security Settings	features/account/account-layout profile-view.ts routes/user.route.js usertoken.route.js	Working: Profile details update, password modification, avatar image uploads, active login session list, and personal notification preferences.	Multi-factor authentication (MFA) setup modal with QR code generation for Google Authenticator.	AI Privacy Guard: Automatically scans profile attachments and document uploads to redact sensitive personal identification numbers.
21	Web Push Notification Engine	features/webpush/components/web-push routes/notifications.route.js notificationqueue.route.js notificationhistory.route.js	Working: Web push subscriber token tracker, push notification composer, broadcast queue, and delivery history logs.	Service worker background push handler for native desktop/mobile OS notification alerts.	AI Optimal Push Scheduler: Analyzes individual user browser activity hours to dispatch web push notifications at peak engagement moments.
22	Email Campaign & SMTP Gateway	features/email/components/email routes/smtpsettings.route.js integrations/email/ smtpsettings.model.js	Working: SMTP connection setup, outbound email campaign composer, HTML message previewer, and sending queue status monitoring.	DKIM, SPF, and DMARC DNS record validator tool to ensure high email deliverability rates.	AI Email Subject Line & Copy Generator: Generates high-converting email subject lines and body copy optimized for inbox deliverability and click-through rates.
23	Knowledge Base RAG & Semantic Vector Search	routes/knowledgebase.route.js services/qdrant.service.js services/kb.service.js knowledgebase.model.js	Working: Article authoring, categories & tags (<code>kbcategories</code> , <code>kbtags</code>), user feedback tracking (<code>kbfeedback</code>), and Qdrant Vector DB embedding sync.	Automatic PDF / Word document parsing and vector chunking when uploading files to the Knowledge Base.	Self-Learning RAG Engine: Automatically identifies unanswered customer questions in chat and drafts new KB articles for team review.

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24	Embedded System Documentation Viewer	features/docs/docs-viewer routes/v1.js	Working: Integrated user manual and API documentation reader within the CRM interface.	In-app search bar filtering documentation pages by keyword or error code.	AI Interactive Docs Assistant: Conversational assistant embedded in docs that answers developer implementation questions instantly with working code examples.

Detailed Technical Execution Flows

Flow 1: Inbound WhatsApp Message → AI RAG / Autopilot → Socket.IO UI Stream

```
sequenceDiagram
    autonumber
    actor Customer
    participant Meta as Meta WhatsApp Cloud API
    participant Webhook as Express Webhook Controller
    participant Queue as BullMQ Redis Queue
    participant Autopilot as Agentic Autopilot Service
    participant Qdrant as Qdrant Vector DB
    participant DB as MySQL DB (Sequelize)
    participant Socket as Socket.IO Server
    actor Agent as CRM Agent UI

    Customer->>Meta: Sends WhatsApp Message
    Meta->>Webhook: POST /v1/webhook/whatsapp
    Webhook->>DB: Store inbound message & update conversation state
    Webhook->>Socket: Emit 'message:received' to project room
    Socket->>Agent: Live Chat UI appends message immediately
    Webhook->>Queue: Push 'process-ai-reply' job
    Queue->>Autopilot: Process incoming message context
    Autopilot->>Qdrant: Query semantic context (Knowledge Base RAG)
    Qdrant->>Autopilot: Return relevant KB snippets
    Autopilot->>Meta: Dispatch AI auto-reply via WhatsApp API
    Autopilot->>DB: Record AI response in database
    Autopilot->>Socket: Emit 'message:sent' (AI Persona)
    Socket->>Agent: Live Chat UI updates with AI response
```

Flow 2: Visual Workflow Node Creation (@foblex/flow) → Backend Execution Engine

```
sequenceDiagram
    autonumber
    actor User as Marketer / Admin
    participant UI as Flow Canvas Component (@foblex/flow)
    participant API as Flows Controller (/v1/flows)
    participant DB as Flows Model
    participant Engine as Flow Execution Engine
    participant Channel as Messaging Service (WhatsApp/Email/SMS)

    User->>UI: Drag-and-drop Trigger, AI, Delay & Message nodes
    User->>UI: Click 'Save & Activate Flow'
    UI->>API: POST /v1/flows (Flow JSON graph metadata)
    API->>DB: Save flow definition & activation status
    Note over Engine: Event Trigger (e.g. New Contact Tagged)
    Engine->>DB: Fetch active flow graph
    Engine->>Engine: Evaluate decision nodes & delays
    Engine->>Channel: Trigger node action (Dispatch WhatsApp / Email message)
    Channel->>DB: Record delivery status
```